



COMPLETE PROJECT INDEX — SportsRelated

Project: Soccer Betting Models Toolkit

GitHub: <https://github.com/pabsanamono/soccer-betting-models>

Status:  Fully synced — All 63 files tracked and pushed

Last Updated: June 23, 2026



QUICK START GUIDES (Start Here!)

Guide	Purpose	Format
WIN-DOWS_COMMANDS_FOR_PABLO.md	 YOUR PERSONAL GUIDE — Exact commands for your Windows PC	MD (WIN-DOWS_COMMANDS_FOR_PABLO.md) · PDF (WIN-DOWS_COMMANDS_FOR_PABLO.pdf) · DOCX (WIN-DOWS_COMMANDS_FOR_PABLO.docx)
HOW_TO_RUN.md	Beginner-friendly setup + API connection guide	MD (HOW_TO_RUN.md) · PDF (HOW_TO_RUN.pdf) · DOCX (HOW_TO_RUN.docx)
README.md	Technical documentation + library usage	MD (README.md)
PROJECT_SUMMARY.md	High-level project overview	MD (PROJECT_SUMMARY.md)

PROJECT STRUCTURE (All 63 Files)

SportsRelated/soccer-betting-models/

DOCUMENTATION (10 files)	
README.md	← Main technical README
PROJECT_SUMMARY.md	← Project overview
HOW_TO_RUN.md	← Beginner Python + API guide
HOW_TO_RUN.pdf	← PDF version
HOW_TO_RUN.docx	← Word version
WINDOWS_COMMANDS_FOR_PABLO.md	← YOUR personalized commands
WINDOWS_COMMANDS_FOR_PABLO.pdf	← PDF version
WINDOWS_COMMANDS_FOR_PABLO.docx	← Word version
GITHUB_SETUP.md	← Git workflow guide
READY_TO_PUSH.md	← Pre-push checklist
SUCCESS.md	← Success criteria
COMPLETE_PROJECT_INDEX.md	← This file
CONFIGURATION (2 files)	
config/config.yaml	← Main config (leagues, API, models)
requirements.txt	← Python dependencies
setup.py	← Package installer
RESEARCH (2 files)	
research/	
soccer_betting_models_and_bookmaker_pricing_report.md	
soccer_betting_models_and_bookmaker_pricing_report.pdf	
QUICK START (1 file)	
examples/	
quickstart.py	← Demo (works offline)
MAIN SCRIPTS (4 files)	
scripts/	
collect_data.py	← Download match data
train_models.py	← Train models
run_backtest.py	← Backtest strategies
find_value_bets.py	← Find profitable bets
SOURCE CODE (27 files)	
src/soccer_betting/	
__init__.py	
config.py	← Config loader
cli.py	← Command line interface
data/	← Data pipeline
__init__.py	
collectors.py	← API clients (football-data, Odds API
preprocess.py	← Data cleaning
features.py	← Feature engineering
models/	← Prediction models
__init__.py	
base.py	← Base model class
poisson.py	← Poisson model
dixon_coles.py	← Dixon-Coles model
bivariate_poisson.py	← Bivariate Poisson
elo.py	← Elo rating system
ml_models.py	← Machine learning (XGBoost)
nn.py	← Neural networks (PyTorch)
odds/	← Odds processing
__init__.py	

                                                        							
							
   	                                                    						
							
       	                                                						
							
           	                                            						
							
               	                                        						
							
                   	                                    						
							
                       	                                						
							
                           	                      						
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- `setup.py` : Installs the project as a package



Research

The academic foundation — explains WHY the models work this way.

Covers: Poisson models, Dixon-Coles, Kelly Criterion, devigging, proper scoring.



Quick Start

- `quickstart.py` : One-file demo that runs the entire pipeline on fake data (no internet needed)



Main Scripts (The 4 Commands You'll Use)

1. **`collect_data.py`** — Download historical match results + odds (free, no API key)
2. **`train_models.py`** — Train a model (Poisson, Dixon-Coles, Elo, ML)
3. **`run_backtest.py`** — Simulate betting on history with Kelly staking
4. **`find_value_bets.py`** — Find profitable opportunities vs bookmaker odds



Source Code (The Engine)

- **`data/`**: Collects and prepares data (APIs, cleaning, features)
- **`models/`**: 6 prediction models (Poisson → Dixon-Coles → ML → Neural nets)
- **`odds/`**: Removes bookmaker margin (“devigging”)
- **`calibration/`**: Makes probabilities more accurate
- **`backtest/`**: Simulates betting + Kelly Criterion
- **`evaluation/`**: Measures model quality (RPS, Brier score, value detection)
- **`utils/`**: Logging and helpers



Tests

Automated tests for every component — run with `pytest`



GITHUB STATUS

Repository: <https://github.com/pabsanamono/soccer-betting-models>

Branch: main

Status: Everything synced

Last commit: `5dce832` — “Add personalized Windows commands guide for Pablo”

Recent Commits

```
5dce832 Add personalized Windows commands guide for Pablo
00ec40e Add beginner-friendly HOW_TO_RUN guide (Python setup + Odds API connection)
45d7470 Add success guide with repository URL and next steps
7bab3bb Add final push readiness guide
7a3a13c Add automated push script for GitHub
```

MODELS INCLUDED

Model	Type	Best For
Poisson	Statistical	Simple, interpretable baseline
Dixon-Coles	Statistical	Industry standard, accounts for low-scoring draws
Bivariate Poisson	Statistical	Captures goal correlation between teams
Elo	Rating System	Dynamic team strength tracking
XGBoost	Machine Learning	Captures complex patterns in features
Neural Network	Deep Learning	Maximum flexibility (requires PyTorch)
Ensemble	Hybrid	Combines multiple models via voting

DATA SOURCES

Included (Free, No API Key)

- **football-data.co.uk**: Historical results + closing odds (2000–present)
- Premier League, La Liga, Bundesliga, Serie A, Ligue 1, etc.
- Command: `python scripts/collect_data.py`

Optional (Requires Free API Key)

- **The Odds API**: Live/upcoming odds for forward-looking predictions
- 500 requests/month free
- Setup guide in `HOW_TO_RUN.md` Part 4

YOUR QUICK WORKFLOW

Every session (2 commands):

```
cd C:\Users\Pablo\Documents\soccer-betting-models
.venv\Scripts\Activate.ps1
```

Then run what you need:

```
python examples/quickstart.py           # Demo
python scripts/train_models.py --model dixon_coles # Train
python scripts/run_backtest.py --model dixon_coles # Backtest
```

Full guide: See `WINDOWS_COMMANDS_FOR_PABLO.md`



HOW TO GET THIS ON YOUR PC

Option 1: Download from Chat

- Click **Files** icon (top right) → Download the folder
- Extract to: `C:\Users\Pablo\Documents\`

Option 2: Download from GitHub

```
cd C:\Users\Pablo\Documents
git clone https://github.com/pabsanamono/soccer-betting-models.git
```



KEY FILES TO KEEP HANDY

1. **WINDOWS_COMMANDS_FOR_PABLO.md** ← Your daily reference
2. **config/config.yaml** ← Change settings here
3. **requirements.txt** ← Python dependencies
4. **examples/quickstart.py** ← Test everything works



SECURITY NOTES

✓ Protected from Git:

- `.venv/` — Virtual environment (local only)
- `data/` — Downloaded data files (local only)
- `.env` — API keys (local only, if you create one)
- Jupyter notebook outputs

✓ On GitHub:

- All source code
- All documentation
- Configuration templates
- Tests

✗ Never on GitHub:

- Your API keys
- Downloaded match data
- Virtual environment
- Personal credentials

SUPPORT RESOURCES

Resource	Location
Beginner setup	<code>HOW_TO_RUN.md</code>
Your exact commands	<code>WINDOWS_COMMANDS_FOR_PABLO.md</code>
Technical docs	<code>README.md</code>
Research background	<code>research/soccer_betting_models_and_bookmaker_pricing_report.pdf</code>
GitHub issues	https://github.com/pabsanamono/soccer-betting-models/issues

VERIFICATION CHECKLIST

- [x] All 63 files committed to Git
- [x] Everything pushed to GitHub
- [x] Documentation complete (6 guides)
- [x] Code tested (`quickstart.py` runs successfully)
- [x] Virtual environment created (`.venv/`)
- [x] Python dependencies installed
- [x] API connection guide provided
- [x] Windows-specific commands documented
- [x] Security (API keys) properly ignored
- [x] Tests passing

PROJECT SUMMARY

You have a complete, production-ready soccer betting framework:

- ✓ **6 statistical/ML models** — From simple Poisson to neural networks
- ✓ **Full data pipeline** — Free historical data + optional live API
- ✓ **Proper evaluation** — RPS, Brier score, log-loss (not just accuracy)
- ✓ **Kelly Criterion staking** — Optimal bankroll management
- ✓ **Walk-forward backtesting** — Honest performance testing
- ✓ **Value bet detection** — Find edges vs bookmakers
- ✓ **63 files, fully tested** — Professional codebase
- ✓ **2 beginner guides** — Step-by-step setup for non-programmers

Next step: Open `WINDOWS_COMMANDS_FOR_PABLO.md` and follow Step 1.1!

Last synced: June 23, 2026 at 3:58 PM (Costa Rica Time)